

Rishab P Nambiar

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Data Scientist with end-to-end experience building, training, and deploying AI models across NLP, Computer Vision, and Reinforcement Learning. Skilled in model development (PyTorch, Scikit-learn, XGBoost), transformer fine-tuning (BERT, RoBERTa), generative modeling (GANs), and Explainable AI (SHAP, LIME, Grad-CAM). Published researcher (ICAISC-2025) and hackathon winner with proven ability to deliver production-ready, metric-driven ML solutions on real-world datasets.

EDUCATION

Amrita Vishwa Vidyapeetham

Integrated M.Sc. in Data Science (4th Year of 5) — GPA: 8.99/10

Coimbatore, India

Sept 2022 – Present

- Relevant Coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Mining, Computer Vision, Natural Language Processing, Reinforcement Learning, Database Systems

Kendriya Vidyalaya No. 1, Calicut

AISSCE (Class XII) — 95.2% | Physics, Chemistry, Mathematics, Biology

2020 – 2021

TECHNICAL SKILLS

ML & Deep Learning Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers

Model Architectures: CNNs, RNNs, BiLSTM, GANs (Pix2Pix), BERT, RoBERTa, DQN, GNN, Multimodal Fusion Networks

Data Science & Analysis: Feature Engineering, Statistical Analysis, Hypothesis Testing, Time-Series Forecasting, A/B Testing concepts

Natural Language Processing: Text Classification, Transformer Fine-tuning, TF-IDF, Word Embeddings, Tokenization

Computer Vision: Object Detection (YOLOv4), Image Classification, GAN-based Image Generation, Grad-CAM

Explainable AI (XAI): SHAP, LIME, Grad-CAM

Programming: Python, R, SQL, MATLAB

Data & Visualization: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, Tableau

Deployment & MLOps: Streamlit, Gradio, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

MultiMalNet: Multimodal Malware Classification

CNN | GNN | Multimodal Fusion | PyTorch

- Architected a trimodal deep learning fusion model combining CNN (visual features), Graph Neural Network (structural features), and Semantic Attribute embeddings to classify malware families from binary image representations.
- Achieved **99.65% accuracy**, **F1-weighted: 99.64%**, and **F1-macro: 99.01%**, outperforming all unimodal baselines; demonstrated the value of multimodal feature fusion for cybersecurity ML applications.

DetectAI: Human vs. AI-Generated Text Classifier

NLP | RoBERTa | BiLSTM | SHAP | Gradio

- Designed a hybrid classification pipeline combining TF-IDF + Random Forest + BiLSTM with a fine-tuned RoBERTa transformer; achieved **98.64% accuracy** across a multi-source benchmark dataset.
- Integrated SHAP and LIME for word-level model interpretability and deployed a Gradio demo for real-time inference, demonstrating full ML pipeline ownership from training through production deployment.

Illegal Fence Detection for Wildlife Conservation

*YOLOv4 | Computer Vision | **Published – ICAISC-2025***

- Contributed to a team research project developing a drone-based YOLOv4 object detection system to identify illegal electric fences in wildlife corridors, supporting data preparation, experimentation, and documentation.
- Co-authored research paper accepted and presented at the **International Conference on Artificial Intelligence and Soft Computing (ICAISC-2025)**.

Image Colorization using Generative Adversarial Networks

GANs | Pix2Pix | PyTorch | Computer Vision

- Built an end-to-end Pix2Pix conditional GAN pipeline for grayscale-to-color image synthesis, achieving **PSNR: 24.56 dB** and **SSIM: 0.899**; implemented custom loss functions combining adversarial and L1 reconstruction loss.
- Established reproducible preprocessing and evaluation workflows enabling systematic model comparison across GAN variants.

Adaptive Traffic Signal Controller via Reinforcement Learning

DQN | Reinforcement Learning | PyGame | Python

- Trained a Deep Q-Network (DQN) agent to learn optimal traffic signal phase-switching policies in a simulated multi-lane intersection environment, reducing average vehicle congestion by **~10%** vs. fixed-time baseline.
- Engineered a custom PyGame simulation as the RL training environment; implemented experience replay and target network stabilization for stable Q-value convergence.

Predictive Demand Modeling – Logistics Industry

Hackathon Winner | *XGBoost | Feature Engineering | Time-Series*

- Part of a winning team at the **SMILE Logistics Innovation Hackathon**; contributed to building an XGBoost-based grocery demand forecasting pipeline with temporal and lag feature engineering.

- Supported model evaluation and cross-validation with time-series splits to ensure robust out-of-sample generalization.

MockViewAI: AI-Powered Mock Interview Platform

LLMs | NLP | Streamlit | Python

- Built a fully functional AI interview coaching system that parses resumes and job descriptions to generate tailored, role-specific interview questions using LLM-based pipelines.
- Implemented automated answer evaluation against structured rubrics with actionable feedback reports; delivered as an end-to-end deployable application.

RESEARCH & PUBLICATIONS

Illegal Electric Fence Detection Using YOLOv4 for Wildlife Conservation

International Conference on Artificial Intelligence and Soft Computing (ICAISC-2025)

Proposed a drone-assisted computer vision detection framework using YOLOv4; paper peer-reviewed, accepted, and presented at ICAISC-2025.

ACHIEVEMENTS & LEADERSHIP

Hackathon Winner: 1st Place, SMILe Logistics Innovation Hackathon – XGBoost demand forecasting for grocery supply chain

Leadership: Technical Team Member, Ideathon '24 at Anokha Tech Fest, Amrita Vishwa Vidyapeetham

Cultural: 4th & 5th Prize, AIU Inter-University Youth Festival South Zone, Group Song (38th & 39th Editions)

CERTIFICATIONS

Python (Basic & Intermediate) — HackerRank

Problem Solving (Basic) — HackerRank